



AI & ML



PROJECT PRESENTATION

Stroke Prediction Using Machine
Learning: From EDA to Deployment

Cologne, Germany

2025



Stroke.ai

Decision Making
based on data insights

Predictions
and Analytics



Scan me!

What?



Stroke.ai is a predictive tool that estimates an individual's probability of experiencing a stroke in the future, it is not a diagnostic tool. we aim to provide risk probabilities, not medical conclusions.

How?



- The model is trained on data from over 5,000 patients, using accessible health features such as age, BMI, blood glucose levels, hypertension, heart disease, and lifestyle factors.
- These inputs feed a machine-learning model that outputs a personalized risk score.

Why?



- Stroke remains one of the leading causes of death and disability worldwide.
- Early risk estimation enables preventive care, targeted lifestyle changes, and timely medical follow-up.
- It is more effective and less costly to predict and prevent early than late-stage diagnosis.
- The goal is not to replace medical diagnosis, but to support awareness and early prevention by highlighting potential risk patterns.



Stroke.ai

Stroke.ai's Dataset Overview

- **Dataset:** Kaggle Stroke Prediction Dataset
- **Rows:** 5,110
- **Target:** stroke

is it a essential variable?

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	NaN	never smoked	1
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1

Is there a correlation between being married and the possibility of having a stroke? or is this a misleading variable?

Has NaNs, to clean data these are gonna be replace by the median

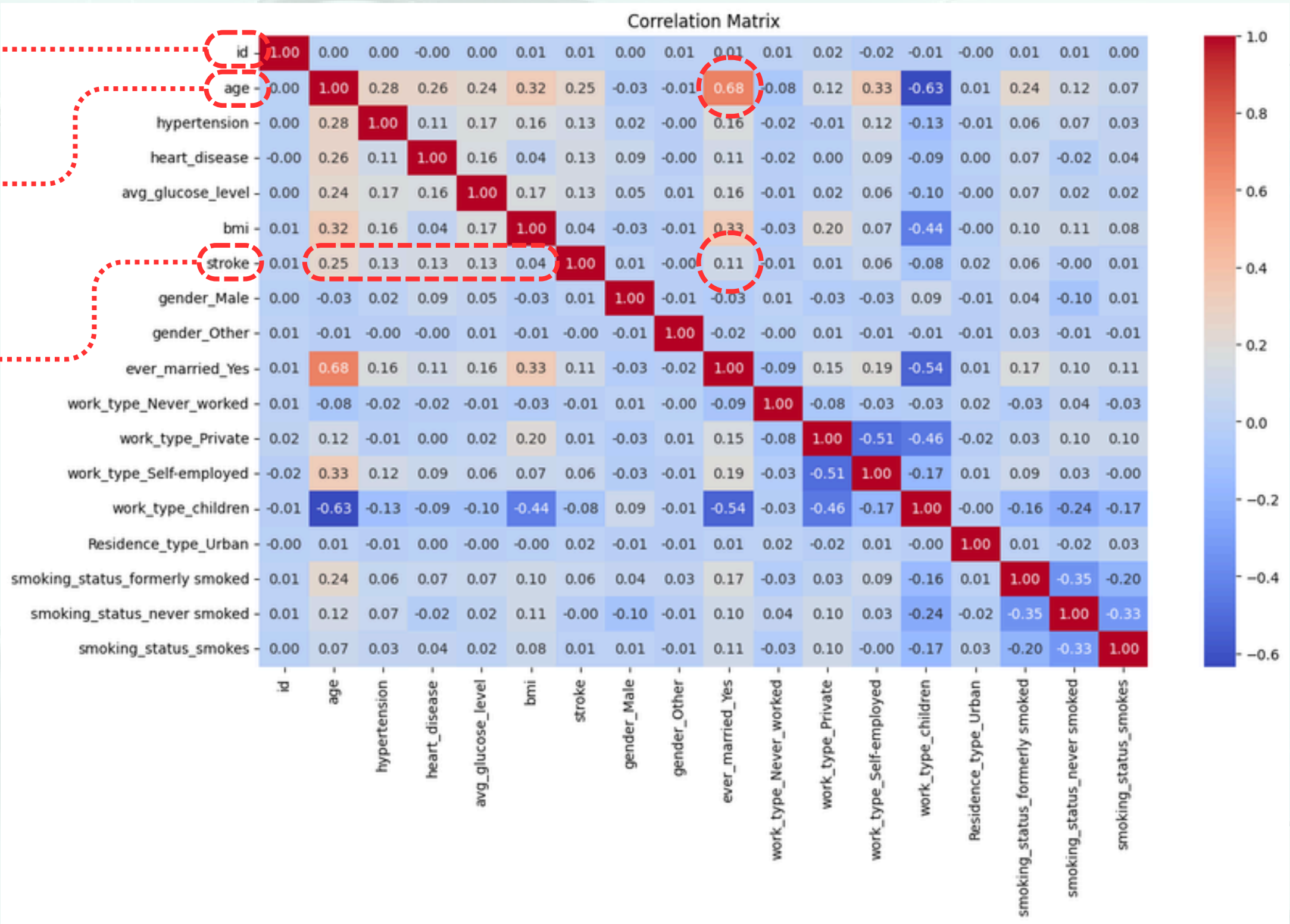


Stroke.ai's Dataset Overview

Non essential variable,
can be dropped

Age is a cofounding variable for both
marital status and stroke, marital status
could be misleading when predicting the
target variable

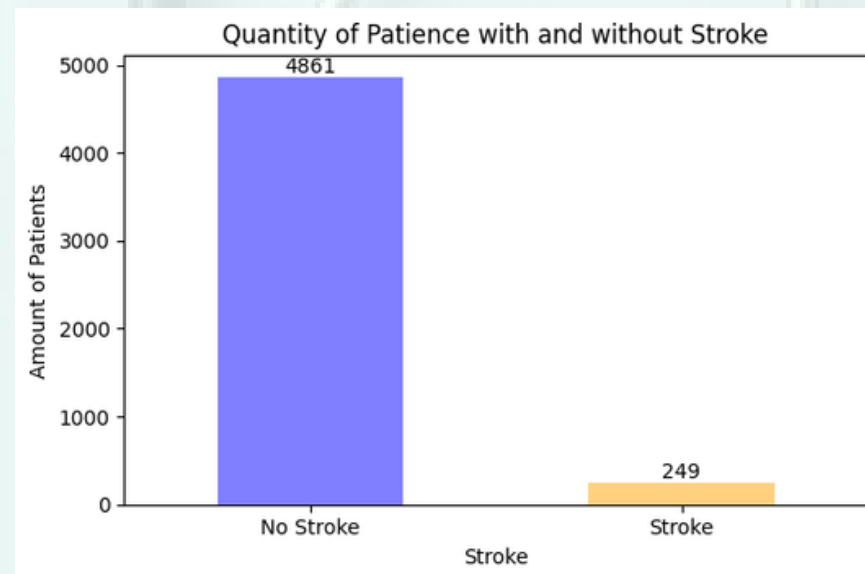
the main features that have a strong
correlation with our target variable
should have a deeper EDA.



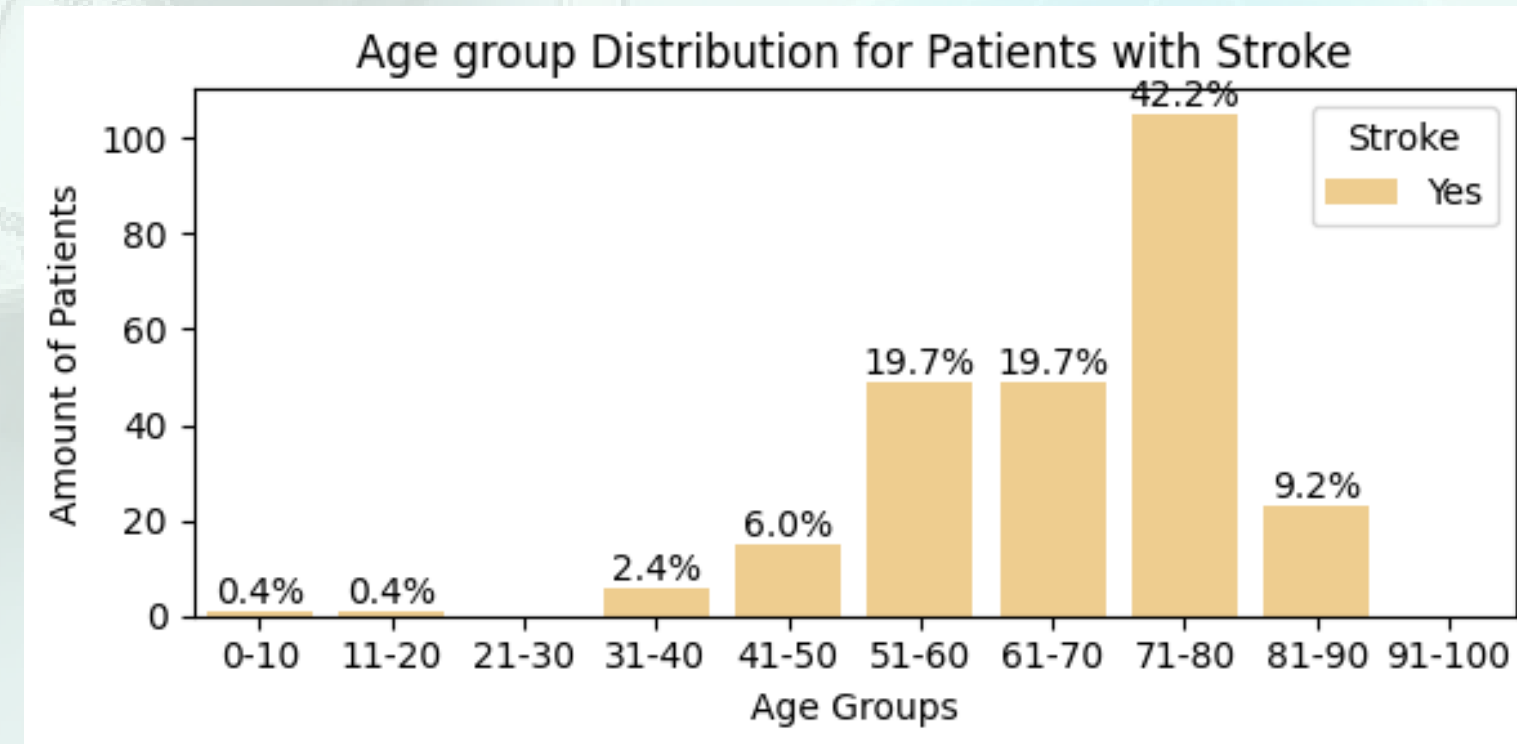
Stroke.ai

Stroke.ai's key features

Age and its role as a cofounding variable



- **Imbalanced dataset:** out of the 5,110 observations, only 5% have had a stroke.



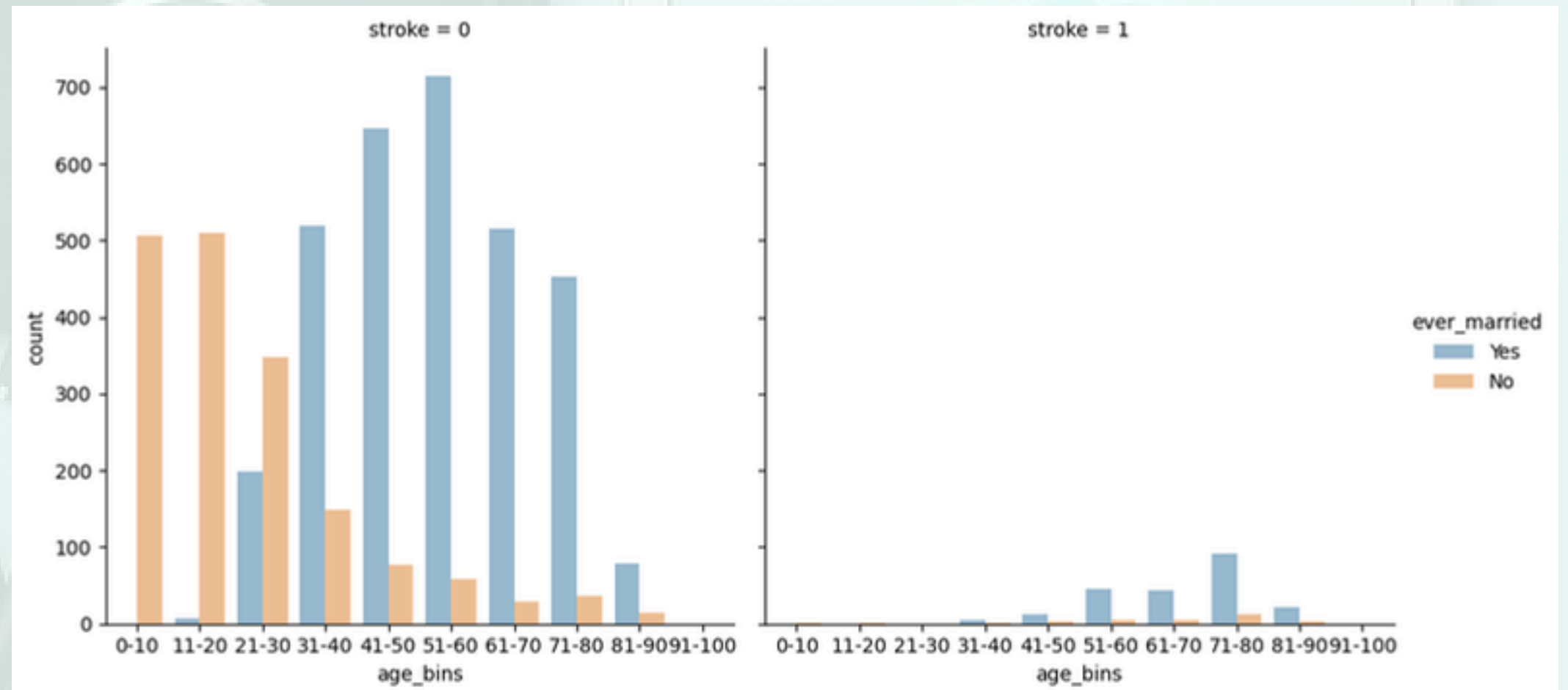
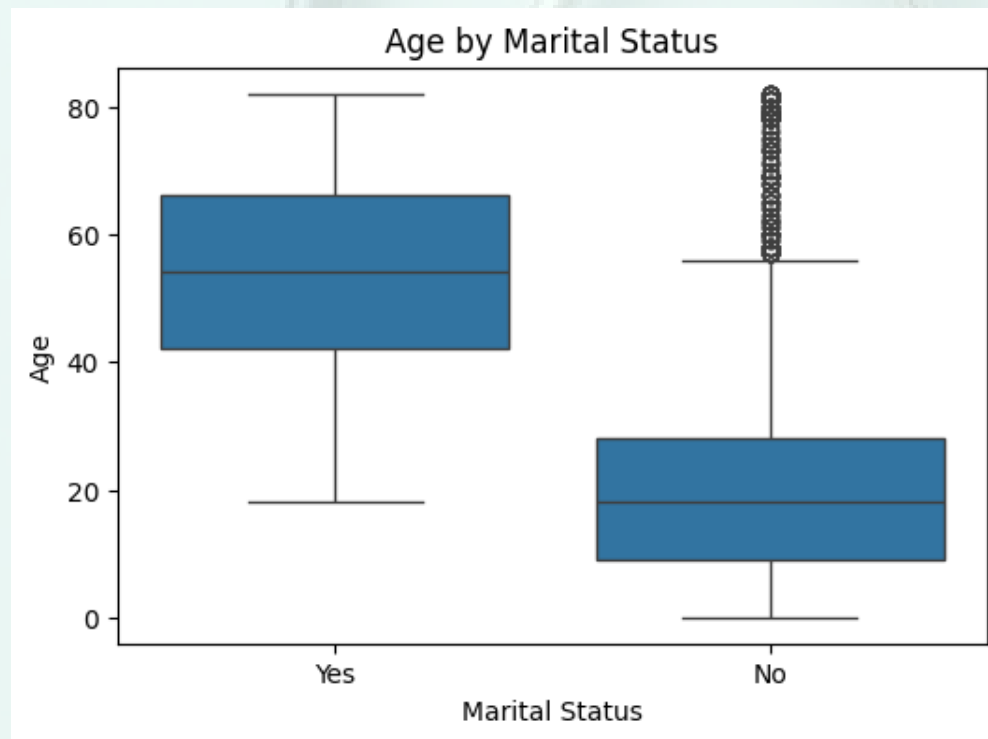
- Age is right-skewed; older individuals dominate stroke cases, with a big jump for individual 71 years and older



Stroke.ai

Stroke.ai's key features

Age and its role as a cofounding variable



- Age is the strongest predictor, but the marital status appear correlated to the target due to confounding.
- Being **married** is then a cause of having a stroke?
- Ever_married importance drops when we include age in the analysis.
- We drop “ever_married” from the final model because its association with stroke is being explain by age, making it a potentially misleading variable.

No!



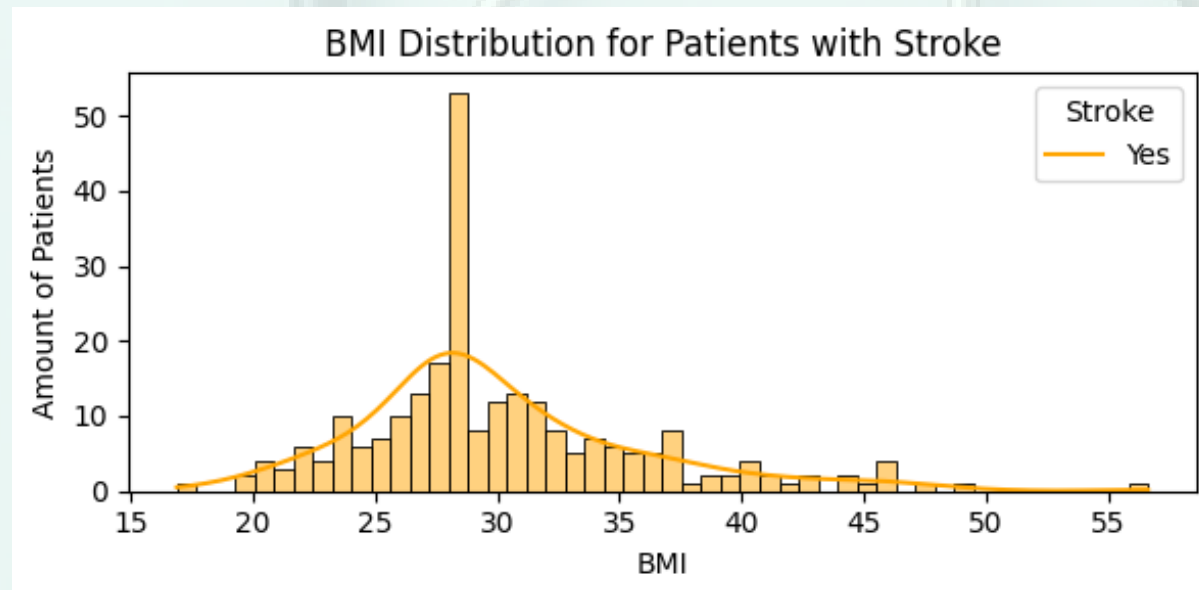
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Ever Married?	Yes	No
Mean	54.3	22.0
Median	54	18
Max	82	82
Min	18	0.1

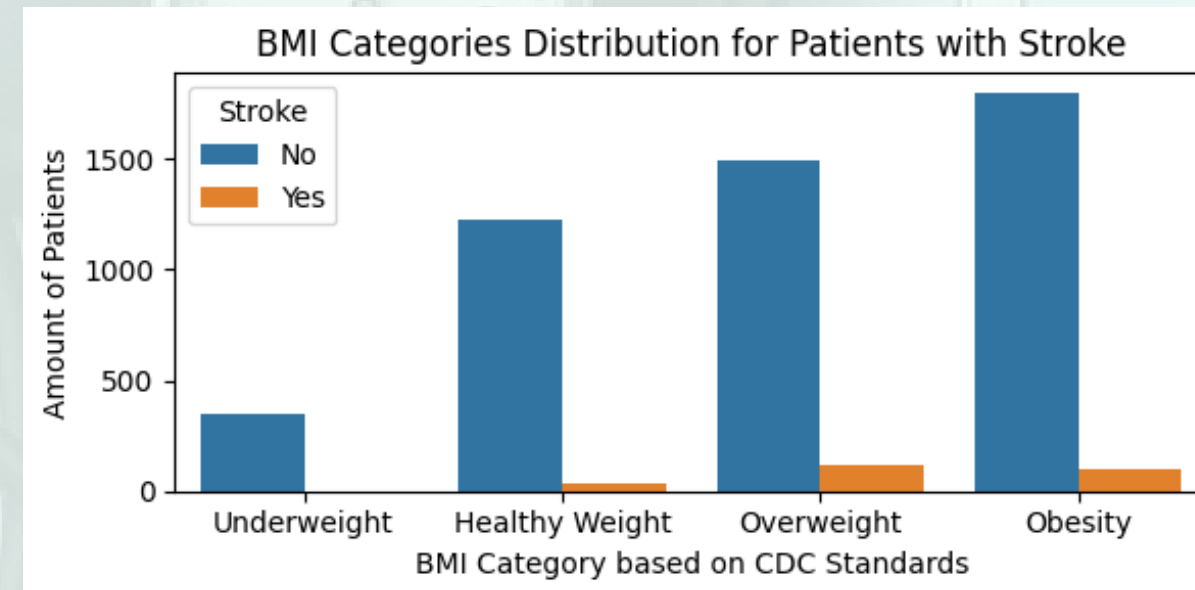
Stroke.ai's key features

BMI is it useful for our model?

BMI provides useful variance to the model because it reflects lifestyle and behavioral health factors.



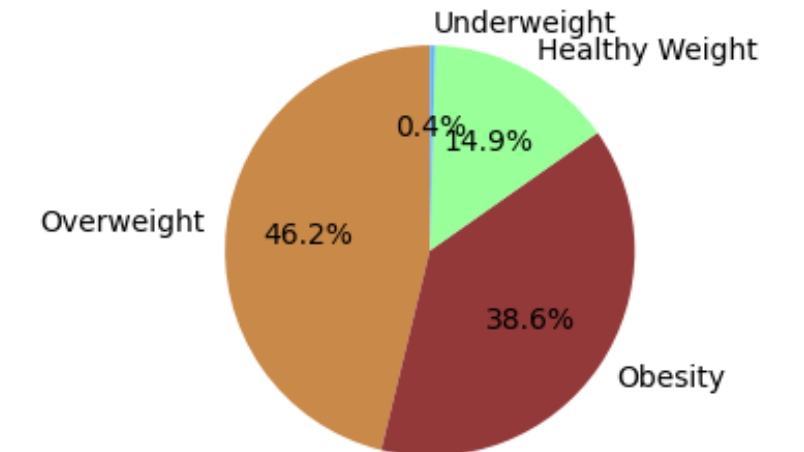
- Visible outliers likely represent individuals with severe obesity or measurement inconsistencies.
- There are missing BMI values, which is expected because BMI often depends on weight/height records.



BMI's bins

$(-\infty, 18.5]$ = Underweight
 $(18.5, 25]$ = Healthy Weight
 $(25, 30]$ = Overweight
 $(30, \infty)$ = Obesity

Proportion of BMI Categories for Patients with Stroke



Despite missing values, the general pattern suggests that higher BMI could contribute to increased stroke risk, but the distribution alone doesn't prove causality

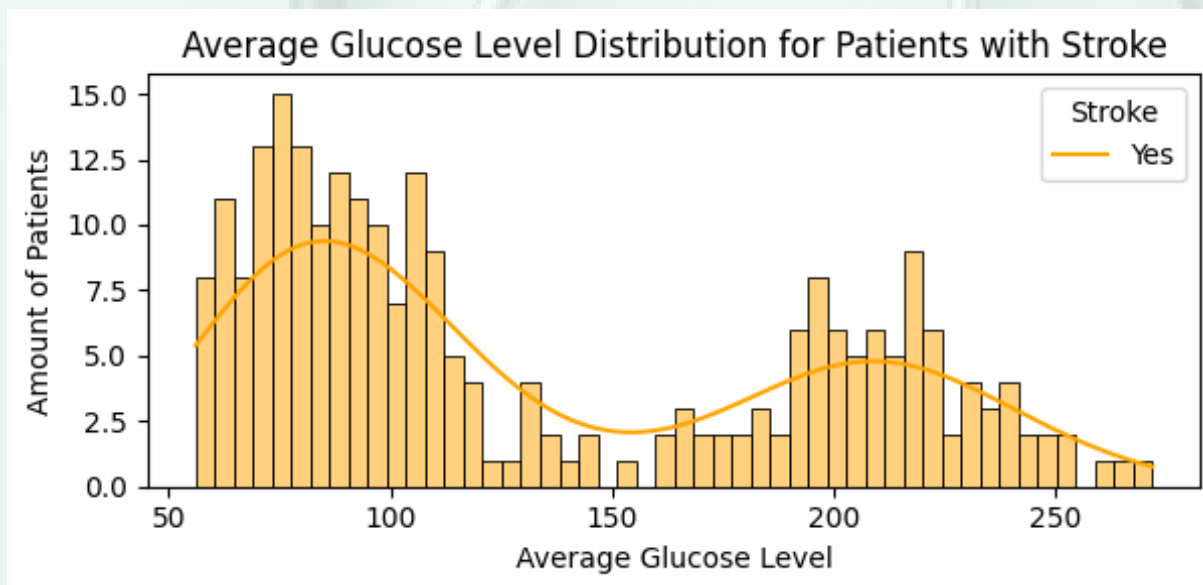


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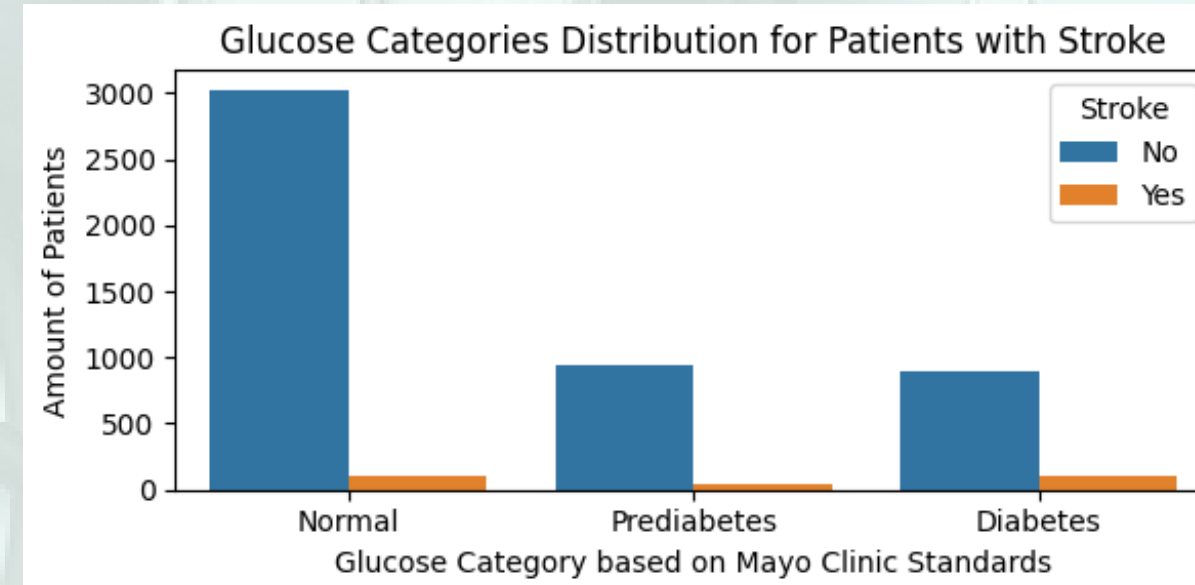
Stroke.ai's key features

Glucose Level is it useful for our model?

glucose correlates with metabolic conditions, making it a predictive and clinically meaningful feature in the model.



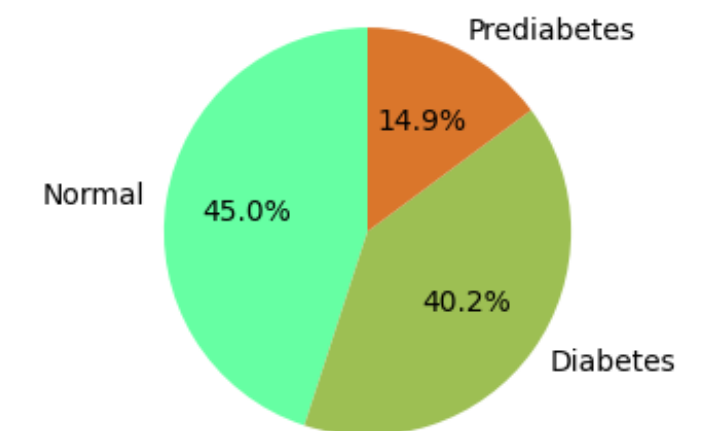
- Although glucose varies widely, no extreme low-value errors appear, suggesting good data quality



Glucose Levels' bins

$(-\infty, 100]$ = Normal
 $(100, 125]$ = Prediabetes
 $(125, \infty)$ = Diabetes

Proportion of Glucose Categories for Patients with Stroke



Elevated glucose levels appear more frequently among individuals with stroke in the dataset, this aligns with known risk factors such as diabetes or pre-diabetes.



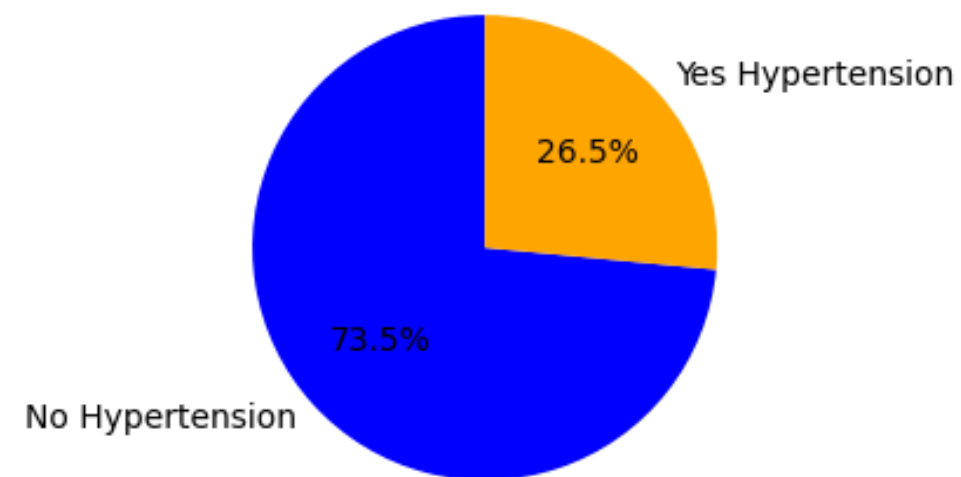
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Stroke.ai's key features

Hypertension and Heart Disease

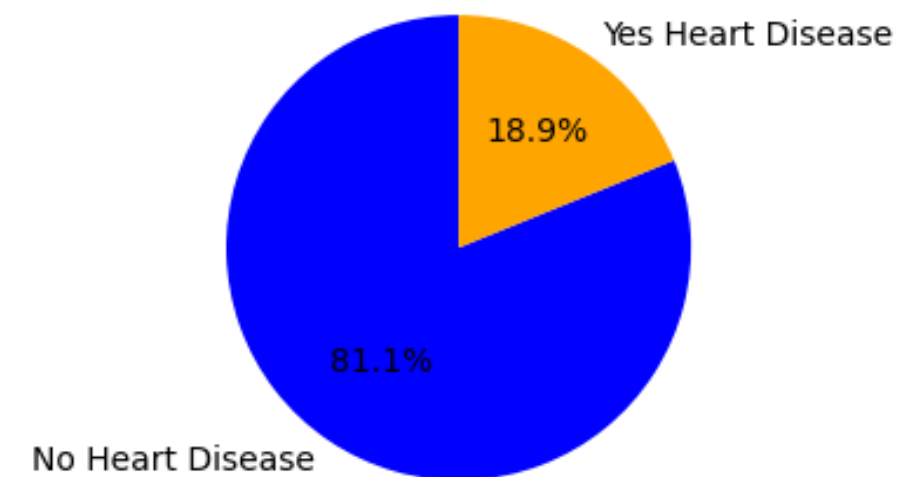
The data imbalance affects the impact of medical conditions in the prediction, but they are still useful for our model

Proportion of Hypertension for Patients with Stroke



- high blood pressure is one of the strongest non-age predictors of stroke.
- Even though the dataset is imbalanced, hypertension remains a high-signal feature

Proportion of Heart Disease for Patients with Stroke



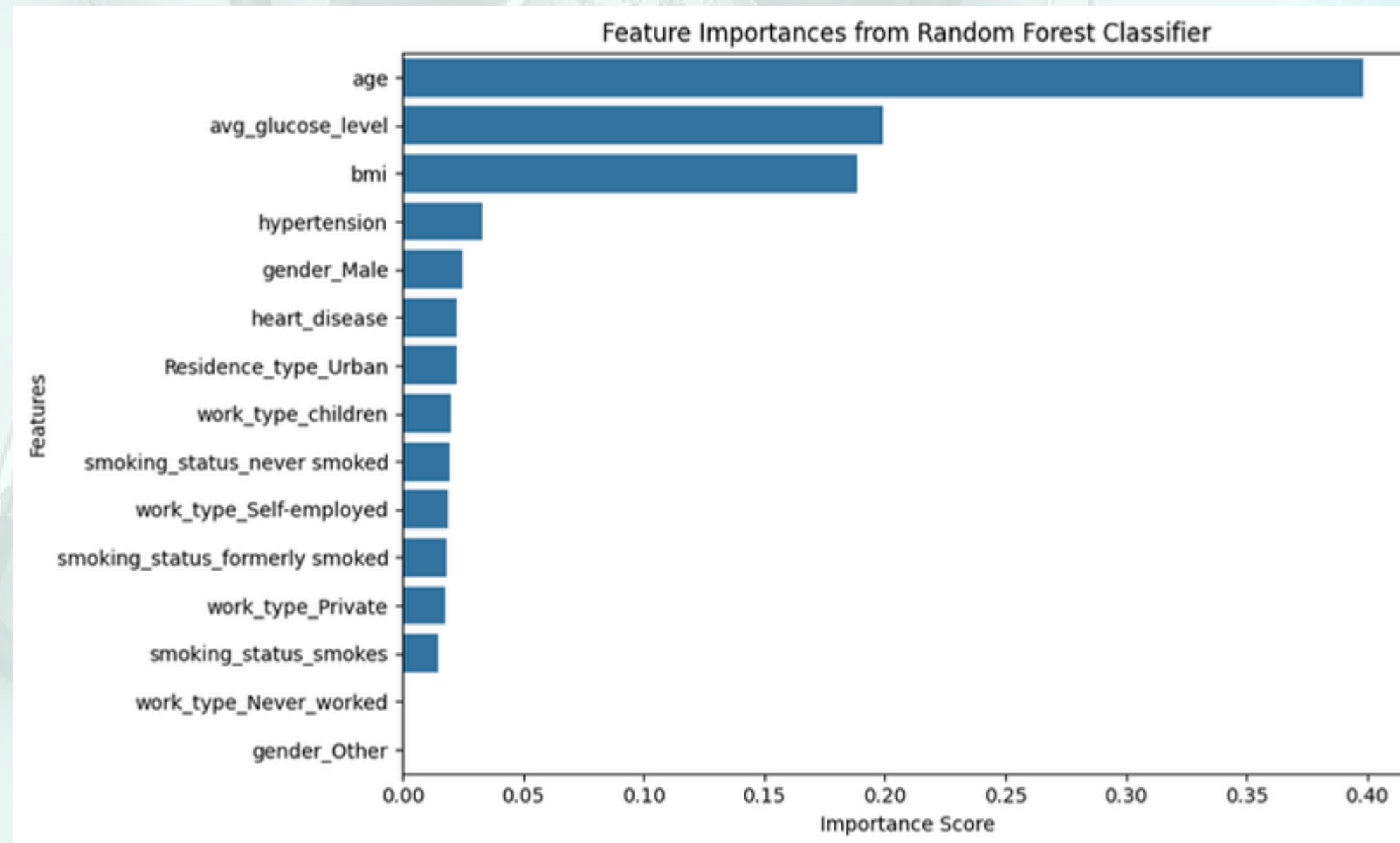
- Heart disease can contribute indirectly to stroke risk through vascular health, but the effect is weaker than hypertension
- The model benefits from it, even if its raw distribution doesn't show dramatic separation.



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Stroke.ai's model

Random Forest Classifier



ever_married was dropped to avoid misleading predictions



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Stroke.ai's model

```
skf = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)
smote = SMOTE(random_state=42)
X_bal, y_bal = smote.fit_resample(x, y)
final_model = RandomForestClassifier(n_estimators=1000,
                                   min_samples_leaf=5,
                                   random_state=42,
                                   class_weight='balanced')
```

Why Random Forest?

- Handles mixed data types: numerical + categorical.
- Captures non-linear relationships between features.
- helps outset the dataset's imbalance.

Why not other models?

- **Logistic regression:** simpler but assumes linear relationships.
- **Decision Tree:** interpretable, but can overfit.

Hyperparameters

- **n_estimators=1000:** more trees, more stable predictions, precision probability by balancing better the weights.
- **class_weight='balanced':** corrects for the class imbalance.
- **min_samples_leaf=5:** smooths probabilities, lowers overfitting and improves recall and precision.

Random Forest Classifier

integrated with K Folds Cross Validation and SMOTE.

- K Folds Cross Validations helps us gain more stable performance predictions by replicating the 80-20 multiple times.
- We stratify to help balance the data and guarantee training is representative.
- SMOTE: Stroke positive cases are rare, we need to oversample to balance the training set and help the model learn patterns. Thus, increasing true positives (but also False Positives).

Is better to have more False positives that with testing can be cleared out, than missing True Positives.



Stroke.ai

Stroke.ai's model

Random Forest Classifier

integrated with K Folds Cross Validation and SMOTE.

Accuracy: 0.9285714285714286

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.97	0.96	972
1	0.21	0.16	0.18	50
accuracy			0.93	1022
macro avg	0.58	0.56	0.57	1022
weighted avg	0.92	0.93	0.92	1022

Initial model without K Folds Cross Validation, Smote, Stratification.

- Precision 21% means that out of all cases predicted as stroke by the model, only 21% are actually stroke cases. This low precision indicates a high rate of false positives.
- Recall 16% means that out of all actual stroke cases, the model correctly identifies only 16% of them. This indicates that the model is missing a 84% of stroke cases, which could be critical in a healthcare context where early detection is vital.

Fold Accuracy: 0.7925636007827789

Fold - Classification Report:

	precision	recall	f1-score	support
0	0.98	0.80	0.88	487
1	0.13	0.58	0.21	24
accuracy			0.79	511
macro avg	0.55	0.69	0.54	511
weighted avg	0.94	0.79	0.85	511

Final model with K Folds Cross Validation, Smote, Stratification.

- Precision decreases to 13%, meaning higher probability of False positives.
- In exchange we get a rate of true positives of almost 60%. proving a better prediction model.

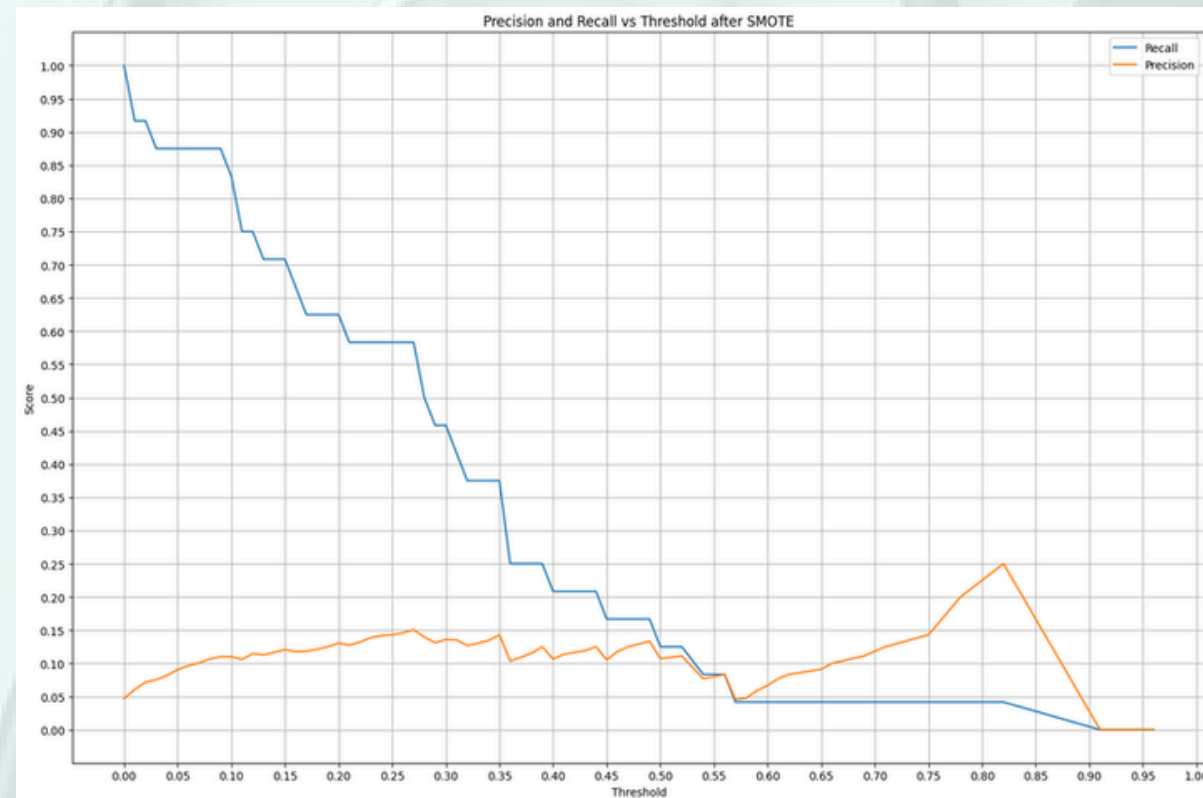


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Stroke.ai's model

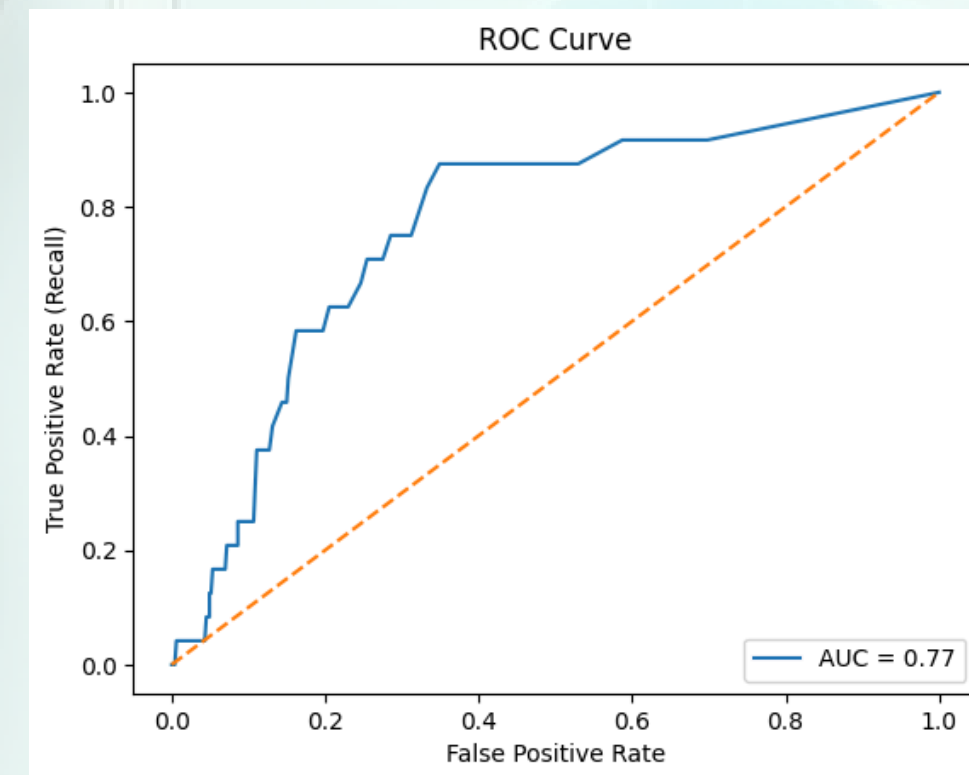
Random Forest Classifier

integrated with K Folds Cross Validation and SMOTE.



This graph help us choose the best possible threshold that gives us a good recall/precision rate. this is needed because Random Forest's default threshold is 50%, not ideal in a medical setting

Chosen Threshold is 20%



What does AUC 77% mean?

The model can correctly distinguish between stroke and non-stroke cases 77% of the time.



Stroke.ai



Stroke.ai

**Predictions
and Analytics**

Q&A

**Decision Making
based on data insights**